

Python Cheat Sheet

Core syntax, data structures, functions, files and quick patterns.

Basics

Topic	Quick reference
Variables	<code>name = "Asha"; age = 22; is_active = True</code>
Data types	<code>int, float, str, bool, list, tuple, dict, set, None</code>
Input/output	<code>name = input("Name: "); print(f"Hi {name}")</code>
Comments	<code># single line comment</code>

Control Flow

Topic	Quick reference
If/elif/else	<code>if score >= 80: ... elif score >= 50: ... else: ...</code>
Loops	<code>for item in items: ... while condition: ...</code>
Loop helpers	<code>break, continue, enumerate(), range(), zip()</code>

Collections

Topic	Quick reference
List	<code>nums=[1,2,3]; nums.append(4); nums[0]; nums[-1]</code>
Dict	<code>user={"name":"Asha"}; user.get("name"); user.keys()</code>
Set	<code>unique=set([1,1,2]); unique.add(3)</code>
Comprehension	<code>squares = [x*x for x in nums if x>0]</code>

Functions & OOP

Topic	Quick reference
Function	<code>def add(a,b=0): return a+b</code>
Lambda	<code>double = lambda x: x*2</code>
Class	<code>class User: def __init__(self,name): self.name=name</code>

Files & Errors

Topic	Quick reference
Read file	<code>with open("file.txt") as f: data=f.read()</code>
Exceptions	<code>try: risky() except ValueError as e: print(e)</code>
Useful modules	<code>os, pathlib, json, datetime, math, random, collections</code>